NLP Part 5: WordNet

What is WordNet?

WordNet is a large lexical database of the English language.

It groups words with similar meanings into sets called **Synsets (Synonym Sets)**.

WordNet also stores relationships between words and concepts.

Example

car → automobile → motorcar

These words have similar meanings and may belong to the same synset.

Why do we use WordNet in NLP?

WordNet can be used for:

- Finding meanings of words
- Finding synonyms
- Finding antonyms
- Finding relationships between words
- Finding more general or specific concepts
- Measuring semantic similarity
- Word sense disambiguation
- Text classification and clustering

In [16]:
```
import nltk

#nltk.download('wordnet')
#nltk.download('omw-1.4')

from nltk.corpus import wordnet as wn

print("WordNet is ready!")
```

```
WordNet is ready!
```

1. Synsets

What is a Synset?

A **Synset** means a **Synonym Set**.

It is a group of words that represent the same or a very similar concept.

A word can have **more than one synset** because the same word can have different meanings.

Example

The word **car** can have different meanings.

WordNet stores these different meanings separately.

In [17]:
```
from nltk.corpus import wordnet as wn

word = "car"

synsets = wn.synsets(word)

print("Synsets of", word)

for synset in synsets:
    print(synset)
```

```
Synsets of car
Synset('car.n.01')
Synset('car.n.02')
Synset('car.n.03')
Synset('car.n.04')
Synset('cable_car.n.01')
```

2. Word Definitions

Each synset has its own **definition**.

This helps us understand the different meanings of a word.

Example

A word such as **bank** can mean:

- A financial institution
- The side of a river

WordNet stores these meanings in different synsets.

In [18]:
```
word = "bank"

for synset in wn.synsets(word):
    print("Synset:", synset.name())
    print("Definition:", synset.definition())
    print()
```

```
Synset: bank.n.01
Definition: sloping land (especially the slope beside a body of water)

Synset: depository_financial_institution.n.01
Definition: a financial institution that accepts deposits and channels the money
into lending activities

Synset: bank.n.03
Definition: a long ridge or pile

Synset: bank.n.04
Definition: an arrangement of similar objects in a row or in tiers

Synset: bank.n.05
Definition: a supply or stock held in reserve for future use (especially in emerg
encies)

Synset: bank.n.06
Definition: the funds held by a gambling house or the dealer in some gambling gam
es

Synset: bank.n.07
Definition: a slope in the turn of a road or track; the outside is higher than th
e inside in order to reduce the effects of centrifugal force

Synset: savings_bank.n.02
Definition: a container (usually with a slot in the top) for keeping money at hom
e

Synset: bank.n.09
Definition: a building in which the business of banking transacted

Synset: bank.n.10
Definition: a flight maneuver; aircraft tips laterally about its longitudinal axi
s (especially in turning)

Synset: bank.v.01
Definition: tip laterally

Synset: bank.v.02
Definition: enclose with a bank

Synset: bank.v.03
Definition: do business with a bank or keep an account at a bank

Synset: bank.v.04
Definition: act as the banker in a game or in gambling

Synset: bank.v.05
Definition: be in the banking business

Synset: deposit.v.02
Definition: put into a bank account

Synset: bank.v.07
Definition: cover with ashes so to control the rate of burning

Synset: trust.v.01
Definition: have confidence or faith in
```

3. Synonyms

What is a Synonym?

A **synonym** is a word that has the same or a similar meaning as another word.

Examples

- car → automobile
- happy → glad
- big → large

WordNet allows us to find synonyms automatically.

In [19]:
```
word = "happy"

synonyms = []

for synset in wn.synsets(word):
    for lemma in synset.lemmas():
        synonyms.append(lemma.name())

print("Word:", word)
print("Synonyms:")

print(set(synonyms))
```

```
Word: happy
Synonyms:
{'well-chosen', 'glad', 'happy', 'felicitous'}
```

4. Antonyms

What is an Antonym?

An **antonym** is a word with the opposite meaning.

Examples

- good ↔ bad
- happy ↔ unhappy
- hot ↔ cold
- large ↔ small

WordNet stores antonym relationships for many words.

In [20]:
```
word = "good"

antonyms = []
```

```
for synset in wn.synsets(word):
    for lemma in synset.lemmas():

        if lemma.antonyms():
            antonyms.append(
                lemma.antonyms()[0].name()
            )

print("Word:", word)
print("Antonyms:", set(antonyms))
```

```
Word: good
Antonyms: {'badness', 'ill', 'evilness', 'evil', 'bad'}
```

5. Synonyms and Antonyms Together

We can combine both operations.

For one word, we can find:

1. Synonyms
2. Antonyms

In [21]:
```
word = "good"

synonyms = set()
antonyms = set()

for synset in wn.synsets(word):

    for lemma in synset.lemmas():

        # Add synonym
        synonyms.add(lemma.name())

        # Add antonym
        if lemma.antonyms():
            antonyms.add(
                lemma.antonyms()[0].name()
            )

print("Original Word:", word)

print("\nSynonyms:")
print(synonyms)

print("\nAntonyms:")
print(antonyms)
```

```
Original Word: good

Synonyms:
{'well', 'dear', 'salutary', 'in_effect', 'adept', 'beneficial', 'ripe', 'good',
'safe', 'full', 'unspoiled', 'secure', 'serious', 'soundly', 'undecomposed', 'ski
llful', 'honest', 'dependable', 'just', 'in_force', 'skilful', 'respectable', 'ef
fective', 'trade_good', 'estimable', 'goodness', 'unspoilt', 'right', 'sound', 'e
xpert', 'commodity', 'honorable', 'practiced', 'proficient', 'near', 'thoroughl
y', 'upright'}

Antonyms:
{'badness', 'ill', 'evilness', 'evil', 'bad'}
```

6. Semantic Relationships in WordNet

WordNet does more than find synonyms.

It also describes relationships between concepts.

Important relationships include:

Relationship	Meaning	Example
Synonym	Similar meaning	car → automobile
Antonym	Opposite meaning	good → bad
Hypernym	More general concept	car → vehicle
Hyponym	More specific concept	car → sports car
Meronym	Part of something	car → wheel

7. Hypernyms

What is a Hypernym?

A **hypernym** is a more general concept.

Example

car → vehicle

A car is a type of vehicle.

Therefore:

vehicle is a hypernym of car.

Another example:

dog → animal

In [22]:
```
car = wn.synset('car.n.01')

print("Word:")
print(car)

print("\nDefinition:")
print(car.definition())

print("\nHypernyms:")
print(car.hypernyms())
```

```
Word:
Synset('car.n.01')

Definition:
a motor vehicle with four wheels; usually propelled by an internal combustion eng
ine

Hypernyms:
[Synset('motor_vehicle.n.01')]
```

In [23]:
```
car = wn.synset('car.n.01')

for hypernym in car.hypernyms():
    print(hypernym.name())
    print(hypernym.definition())
```

```
motor_vehicle.n.01
a self-propelled wheeled vehicle that does not run on rails
```

8. Hyponyms

What is a Hyponym?

A **hyponym** is a more specific concept.

For example:

vehicle ↓ car ↓ sports car

A sports car is a more specific type of car.

Therefore:

sports car is a hyponym of car.

In [ ]:
```
car = wn.synset('car.n.01')

hyponyms = car.hyponyms()

print("Hyponyms of car:")

for word in hyponyms:
    print(word.name())
```

9. Meronyms

What is a Meronym?

A **meronym** represents a part of something.

Example

A car contains:

- wheel
- engine
- door

Therefore, a **wheel is part of a car**.

This is called a **part-of relationship**.

In [24]:
```
car = wn.synset('car.n.01')

parts = car.part_meronyms()

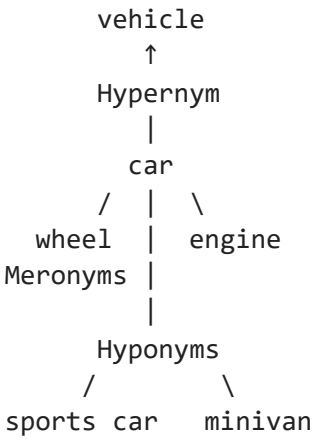
print("Parts of a car:")

for part in parts:
    print(part.name())
```

```
Parts of a car:
buffer.n.06
fender.n.01
floorboard.n.02
glove_compartment.n.01
car_window.n.01
grille.n.02
accelerator.n.01
first_gear.n.01
car_mirror.n.01
stabilizer_bar.n.01
car_door.n.01
reverse.n.02
bumper.n.02
car_seat.n.01
high_gear.n.01
third_gear.n.01
window.n.02
tail_fin.n.02
running_board.n.01
air_bag.n.01
hood.n.09
luggage_compartment.n.01
automobile_engine.n.01
roof.n.02
gasoline_engine.n.01
auto_accessory.n.01
sunroof.n.01
automobile_horn.n.01
rear_window.n.01
```

WordNet Relationship Summary

Consider the word **car**:



Remember

Hypernym = More general

car → vehicle

Hyponym = More specific

car → sports car

Meronym = Part of

car → wheel

10. Complete WordNet Example

Now we will combine several WordNet operations.

For a given word, we will display:

1. Synsets
2. Definition
3. Synonyms
4. Hypernyms
5. Hyponyms

In [25]:
```
from nltk.corpus import wordnet as wn

word = "dog"

synsets = wn.synsets(word)

print("WORD:", word)
print("=" * 50)

# Show available synsets
print("\nSYNSETS:")

for synset in synsets:
    print(synset.name())

# Select first meaning
first_synset = synsets[0]

print("\nSELECTED SYNSET:")
print(first_synset.name())

print("\nDEFINITION:")
print(first_synset.definition())

print("\nSYNONYMS:")

for lemma in first_synset.lemmas():
    print(lemma.name())

print("\nHYPERNYMS:")

for hypernym in first_synset.hypernyms():
    print(hypernym.name())

print("\nHYPONYMS:")
```

```
for hyponym in first_synset.hyponyms():
    print(hyponym.name())
```

```
WORD: dog
==================================================

SYNSETS:
dog.n.01
frump.n.01
dog.n.03
cad.n.01
frank.n.02
pawl.n.01
andiron.n.01
chase.v.01

SELECTED SYNSET:
dog.n.01

DEFINITION:
a member of the genus Canis (probably descended from the common wolf) that has be
en domesticated by man since prehistoric times; occurs in many breeds

SYNONYMS:
dog
domestic_dog
Canis_familiaris

HYPERNYMS:
domestic_animal.n.01
canine.n.02

HYPONYMS:
spitz.n.01
basenji.n.01
great_pyrenees.n.01
working_dog.n.01
poodle.n.01
toy_dog.n.01
mexican_hairless.n.01
puppy.n.01
hunting_dog.n.01
newfoundland.n.01
griffon.n.02
corgi.n.01
pooch.n.01
dalmatian.n.02
leonberg.n.01
cur.n.01
pug.n.01
lapdog.n.01
```

Exercise: Explore WordNet

Choose one of the following words:

- dog
- car

- computer
- teacher
- student

For your selected word, find:

1. Synsets
2. Definition
3. Synonyms
4. Antonyms (if available)
5. Hypernyms
6. Hyponyms

Question

What did you learn about the relationship between your word and other words?

In [ ]:
```
from nltk.corpus import wordnet as wn

word = "computer"

# Find synsets
synsets = wn.synsets(word)

print("Word:", word)

print("\nSynsets:")
for synset in synsets:
    print(synset)

# Select first synset
selected = synsets[0]

print("\nDefinition:")
print(selected.definition())

print("\nSynonyms:")
for lemma in selected.lemmas():
    print(lemma.name())

print("\nHypernyms:")
for item in selected.hypernyms():
    print(item.name())

print("\nHyponyms:")
for item in selected.hyponyms():
    print(item.name())
```

WordNet Summary

In this lesson, we learned that **WordNet** is a lexical database used in NLP.

We learned the following concepts:

- **Synset** → Group of words with similar meanings
- **Synonym** → Similar meaning
- **Antonym** → Opposite meaning
- **Hypernym** → More general concept
- **Hyponym** → More specific concept
- **Meronym** → Part-of relationship

WordNet can be used in many NLP applications such as:

- Word sense disambiguation
- Semantic similarity
- Information retrieval
- Text classification
- Text clustering
- Query expansion